



Invited review: Defining cattle handler skill— A systematic review and conceptual framework

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ABSTRACT

Cattle handling is a common form of human–animal interaction, is essential to nearly all husbandry practices in cattle production, and has direct implications for animal welfare, productivity, and handler safety. Despite its critical role, the human skill involved in handling remains poorly defined, limiting efforts to evaluate stockperson performance, develop effective training programs, and improve outcomes for animals and workers. Defining handler skill is also essential for understanding how human–animal interactions influence animal behavior and welfare. To address this gap, we conducted a systematic review of peer-reviewed literature focused on cattle handling and stockperson factors. Articles were retrieved from the Web of Science database and screened using predefined inclusion criteria, including a focus on cattle, handler influence, and empirical data. Of the records initially identified, 57 met the inclusion criteria. Using grounded theory methods, data were extracted and synthesized into a handler-centered taxonomy of cattle handling skill, termed the VITAL model (Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Continuous Learning). The VITAL model describes 5 interrelated dimensions of handler skill that together capture both observable handling actions and the perceptual and cognitive components that support skilled practice. This taxonomy provides a comprehensive, evidence-informed framework for defining and evaluating cattle handler competence. It can support the development of training curricula, assessment tools, and behavior-based welfare standards. By providing a shared vocabulary for handler skill, this work contributes to more consistent training outcomes and to improvements in animal welfare, worker safety, and productivity across dairy and beef production systems.

Key words: cattle handling, handler skill, stockmanship, competency framework

INTRODUCTION

“NOTICE TO THE HELP: The rule to be observed in this stable at all times, toward the cattle, young and old, is that of patience and kindness. A man’s usefulness in a herd ceases at once when he loses his temper and bestows rough usage. Men must be patient. Cattle are not reasoning beings. Remember that this is the Home of Mothers. Treat each cow as a Mother should be treated. The giving of milk is a function of Motherhood; rough treatment lessens the flow. That injures me as well as the cow. Always keep these ideas in mind in dealing with my cattle.” ~W. D. Hoard

This guidance, printed in *Hoard’s Dairyman* in the early 20th century, reflects a longstanding cultural understanding within the dairy industry: that cattle handling is not only a technical task, but a relational and ethical one. Although such values have been passed through generations of producers and practitioners, they have not been fully translated into the scientific literature. This review builds on that foundation to examine how handler skill has been conceptualized and to propose a structured taxonomy for identifying and describing its key dimensions.

Handling represents one of the most critical points of contact between people and cattle, directly influencing animal health, productivity, and welfare (Schmied et al., 2008), and worker safety. When handling is rushed or poorly executed, cattle exhibit stronger behavioral stress responses, the risk of injury increases for both humans and animals, and productivity can decline (Grandin, 1993; Andrade et al., 2001; Hemsworth et al., 2011). In contrast, skilled handling promotes calm cattle, safer working environments, and greater public confidence in the industry (Rushen et al., 1999; Grandin, 2003; Woiwode and Grandin, 2014).

Despite its central role in cattle care, handler skill has not been clearly defined. Most existing studies have focused on isolated elements such as attitude, loosely

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

defined low-stress techniques, or facility design, rather than viewing skill as an integrated set of human abilities expressed in practice. Earlier work has emphasized either stockperson attitudes (Hemsworth et al., 2011; Moggy et al., 2017), knowledge of animal behavior (Waiblinger et al., 2002), or low-stress handling methods (Grandin, 1989). Although each line of inquiry has advanced understanding, these efforts have largely evolved in parallel, producing valuable but fragmented perspectives on how handlers influence cattle outcomes. This review addresses that gap by synthesizing current evidence to conceptualize handler skill as a comprehensive construct.

Welfare frameworks such as the Five Freedoms (Brambell, 1965) and the Five Domains (Mellor and Reid, 1994) were developed to guide the assessment of animal welfare in systems where care is provided by humans. Fraser et al. (1997) proposed the Three Circles model to describe the different ethical concerns people may hold about animals' quality of life, including biological functioning, affective state, and natural living. Although these frameworks acknowledge the importance of human influence on animal outcomes, none systematically incorporate the human dimensions of handling skill. More importantly, there is currently no global or holistic approach for objectively defining, describing, or integrating handler skill into welfare science, despite evidence that human competence is fundamental to achieving good welfare outcomes (Boivin et al., 2007; Yost, et al., 2020; Vieira et al., 2023).

The mutual benefit shared by humans and animals living in close proximity (Rollin, 1990) can be traced back thousands of years to the origins of domestication (Larson and Fuller, 2014). So central were animals to human societies that their importance is recorded in archeological remains, hieroglyphics, song, story, art, and legend, reflecting the profound role animals have played in shaping civilizations, cultures, and social systems across history that coexisted with animals. Within this tradition, husbandry skills, and particularly the ability to handle cattle, have long been taught, shared, and valued outside of formal scientific literature.

The concept of slow knowledge has been used to describe this traditional approach to generational learning (Orr, 1996). Applied to stockmanship, it captures how core competencies such as reading animal behavior, anticipating cause and effect, and responding to dynamic conditions were once acquired gradually through daily interaction with animals (Woiwode, 2015). Mastery required not only repeated practice, but also deep familiarity developed over years of coexistence with livestock. In modern agricultural systems, such immersive learning opportunities are increasingly rare, even as evidence shows that animals value and respond to the relationships formed through consistent, skillful human interaction (Anthony,

2003; Hemsworth, 2003; Zulkifli, 2013; Napolitano et al., 2020). This shift underscores the importance of developing structured frameworks for teaching and supporting handler skill in today's production systems.

The present review addresses the need for a structured understanding of cattle handling skill by synthesizing current research on human factors in livestock systems. The main objective of this review is to develop a comprehensive taxonomy of handler skill that can be used to identify and describe the key dimensions of skilled handling. This taxonomy identifies 5 core aspects of handler skill: values and attitudes, technical execution, integration of knowledge, situational awareness, and commitment to continuous learning as core components of handler performance. By treating handler skill as a combination of identifiable and interrelated abilities expressed in practice, this framework provides a practical foundation for training, communication, and future research, with the ultimate aim of supporting both animal welfare and human safety.

MATERIALS AND METHODS

Search Strategy and Selection Criteria

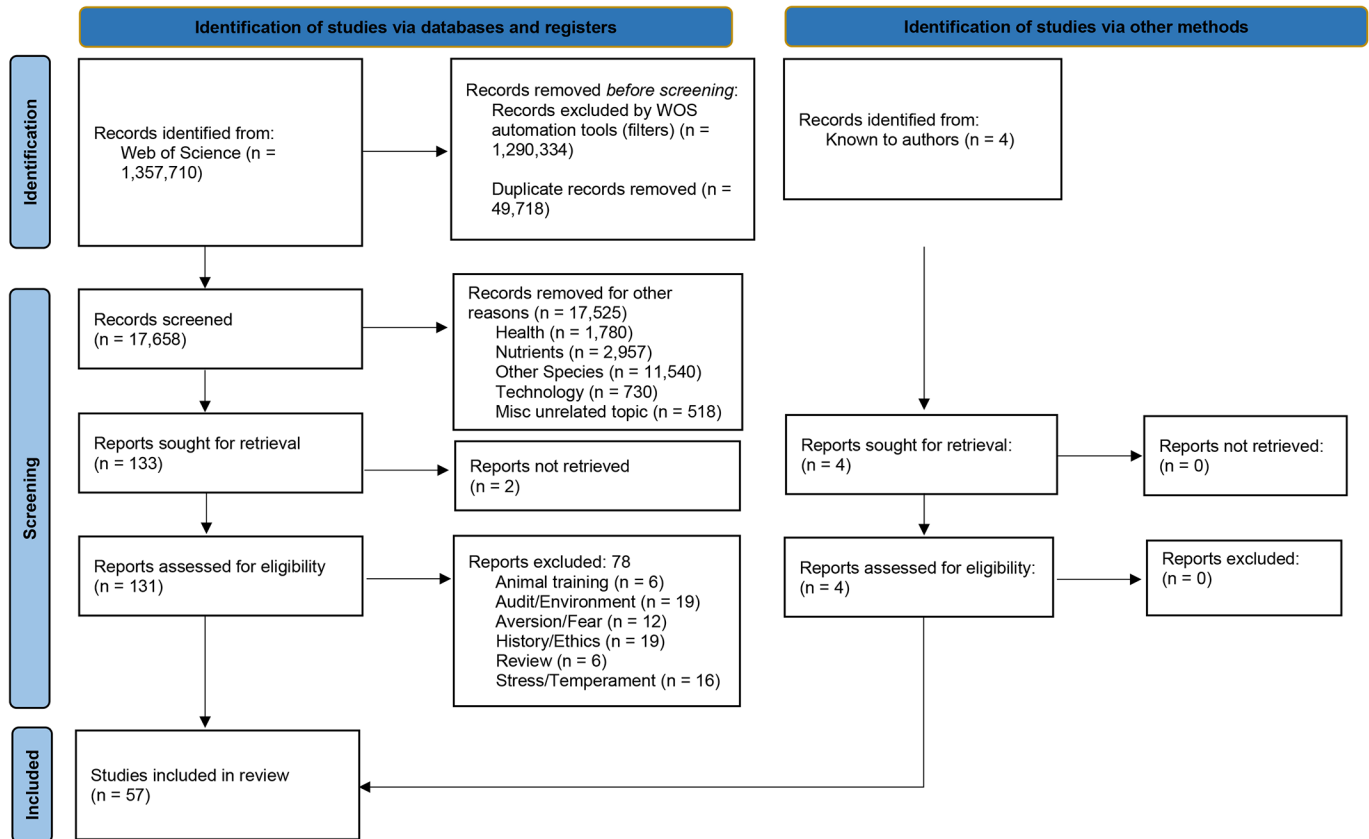
A systematic literature search was conducted in the Web of Science (WOS) Core Collection (<https://www.webofscience.com/wos/>) between September and December 2024 to identify peer-reviewed records examining human characteristics believed to comprise cattle handler skill and their relationship to cattle outcomes. Search queries combined terms such as cattle handling, stockmanship, handler skill, attitude, training, behavior, and welfare. To capture relevant concepts comprehensively, 3,925 unique query combinations were developed (Appendix A). Searches were limited to English-language records, with no date restrictions applied. This review was conducted in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 guidelines for systematic reviews (Page et al., 2021). The completed checklist is provided as supplemental material (see Notes; Woiwode, 2026), and the PRISMA flow diagram (Figure 1) summarizes the screening and inclusion process.

The WOS "Citation Topics Micro," "Citation Topics Meso," and "Categories" filters were used to refine the search, excluding topics outside the scope of this review (e.g., meat quality, non-cattle species). Additional relevant articles known to the authors were also reviewed to supplement the database search.

Records were included if they

1. Examined cattle handling or handler skill in the context of livestock production, welfare, or safety.

PRISMA 2020 flow diagram for new systematic reviews, which included searches of databases, registers, and other sources



Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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Figure 1. The PRISMA 2020 flow diagram illustrating the screening and inclusion process. Source: Page et al. (2021). Misc = miscellaneous.

2. Reported on human attributes (e.g., attitude, training, experience, decision making) or handling practices (e.g., pressure and release, timing, facility use).

Records were excluded if they

1. Focused solely on facility or equipment design without reference to human handling.
2. Addressed species other than cattle (unless multi-species studies provided cattle-specific data).
3. Were not peer-reviewed (e.g., theses, conference abstracts, popular press).
4. Were narrative or conceptual reviews without original empirical data.

Screening and Selection. The initial search yielded 1,357,710 records. After applying WOS filters and removing duplicates, 17,658 unique records remained. To determine relevance for inclusion, the second author and an artificial intelligence (AI) language model (ChatGPT, OpenAI) each independently screened article

titles against the inclusion criteria. These preliminary assessments served to identify potentially relevant records but were not used as final decisions. The corresponding author then reviewed all screening outcomes, verified agreement across reviewers, and made the final determination for inclusion or exclusion of every record. At all stages, human judgment guided the decision-making process, and the use of AI was limited to a supporting role. This process identified 133 records for abstract and full-text review.

Data Extraction and Quality Appraisal. From each of the 133 eligible records, we extracted bibliographic information, study design, cattle production system, handler population, methodological approach, key findings, and reported implications. Data extraction focused specifically on descriptions of handler skills and related human factors, rather than animal outcomes alone.

Study reporting quality and methodological clarity were appraised using the ARRIVE 2.0 guidelines (Percie du Sert et al., 2020), which provided a structured framework to guide extraction and support consistent interpre-

tation across studies published over 6 decades. This process allowed for greater comparability and strengthened the overall rigor of the synthesis.

As part of full-text review and detailed data extraction, additional exclusions were made based on the relevance and completeness of reported methods and findings. This resulted in a final set of 57 peer-reviewed papers included in the synthesis.

Analysis of Included Studies

Extracted information was analyzed using a qualitative approach informed by grounded theory methodology (Glaser and Strauss, 1967; Charmaz, 2014). Grounded theory is a qualitative method in which theoretical insights are developed inductively from data through systematic coding and constant comparison, allowing patterns to emerge from the literature rather than being imposed *a priori*. Coding was performed manually in Microsoft Excel using iterative annotation, color-coding, and sorting to organize and synthesize findings across studies.

An iterative, 3-stage coding process was used to develop the taxonomy of cattle handling skill. During open coding, all explicit references to handler abilities, attributes, and behaviors were labeled and recorded in Excel. Axial coding then grouped similar codes into broader conceptual categories that recurred across production systems and research designs. Finally, selective coding integrated these categories into 5 central dimensions of handler skill, representing the final **VITAL** (Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Continuous Learning) framework.

Language from the original sources was retained where possible to preserve the integrity of reported findings, while also ensuring that the final framework used terms accessible to cattle handlers and training professionals. The complete codebook, including all codes, descriptions, and grouped thematic categories, is available in the supplemental material (see Notes; Woiwode, 2026).

RESULTS

The search identified 57 records that met inclusion criteria. These included quantitative field studies, experimental interventions, surveys, and ethnographic analyses, spanning diverse geographic regions and production systems. Studies examined stockperson training (Austin et al., 2007; Progar et al., 2019), attitudes and empathy (Waiblinger et al., 2002; Panamá Arias and Špinka, 2005; Rault et al., 2020), safety and risk (Lindahl et al., 2012, 2013, 2015; Tone and Irwin, 2022), and facility use and design (Grandin, 1980, 1989).

Emergent Dimensions of Handler Skill

Across the retained records, recurring themes of handler competence were identified and clustered into 5 dimensions of cattle handler skill: Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Commitment to Continuous Learning. These 5 dimensions reflect the complexity of the art and science of cattle handling, from the physical motor skills required to move cattle calmly and safely, to the mental and emotional capacities that guide how handlers think about cattle, how they approach learning about and caring for their animals, down to the nuances of how they read and interpret nonverbal behavioral cues specific to cattle.

Values and Attitudes

Values and attitudes represent the foundational dimension of cattle handling competence, encompassing stockpersons' beliefs about animal sentience, ethical responsibilities toward cattle welfare, and emotional engagement with animals under their care. This dimension reflects the cognitive and affective orientations that shape how handlers perceive their role and responsibilities in livestock management contexts (Waiblinger et al., 2002; Hemsworth, 2003; Rault et al., 2020). Research demonstrates that handler attitudes function as psychological precursors to observable behavior, influencing the quality of human–animal interactions and subsequent animal welfare outcomes (Lensink et al., 2000a,b; Hemsworth et al., 2002; Fukasawa et al., 2017). The values and attitudes dimension extends beyond simple preference to encompass deeply held beliefs about animal stewardship, empathy toward cattle, and commitment to welfare-centered practices (Ellis, 2013; Bassi et al., 2019).

Practical applications of positive values and attitudes manifest in daily handling routines, facility design decisions, and responses to challenging situations. Handlers with positive attitudes toward cattle demonstrate greater patience during routine procedures, invest more effort in gentle handling techniques, and prioritize animal comfort even when time pressures exist (Seabrook and Wilkinson, 2000; Boivin, et al., 2003; Mounier, et al., 2008). Research in dairy operations found that farmers who called cows by name, an indicator of positive attitudes and individualized attention, achieved higher milk yields, suggesting that attitudinal orientations translate into tangible welfare and productivity benefits (Bertenshaw and Rowlinson, 2009). Similarly, beef cattle producers who expressed enjoyment working with animals and felt strong responsibility for animal care were more likely to implement welfare-enhancing management practices (Spooner et al., 2012). These findings illustrate how values and attitudes influ-

ence countless small, daily decisions that collectively determine animal welfare outcomes.

Specific components of this dimension include empathy toward animals, beliefs about animal sentience and emotional capacity, stewardship orientation, and attitudes toward specific husbandry tasks. Empathy, which is the capacity to recognize and respond appropriately to emotional states (including those of animals), has been identified as a critical component linking handler characteristics to interaction quality (Leon, et al., 2020). Studies examining belief in nonhuman animal sentience found that female and older attendees at animal welfare symposia demonstrated stronger emotional engagement with animals, suggesting demographic variation in empathetic orientation (Cornish, et al., 2018). Attitudes toward specific tasks also matter; stockpersons who viewed certain husbandry procedures as unpleasant performed those tasks with less care, highlighting the importance of addressing task-specific attitudes in training programs (Seabrook and Wilkinson, 2000). The stewardship perspective, which frames the handler as a caretaker responsible for animal wellbeing, represents a particularly influential attitudinal orientation in beef production contexts (Ellis, 2013).

Safety and welfare implications of handler values and attitudes are substantial and bidirectional. Positive attitudes reduce cattle fear responses, in turn decreasing the likelihood of defensive or flight behaviors that pose injury risks to both animals and handlers (Petherick et al., 2009a,b; Breuer et al., 2000). Conversely, negative attitudes predict increased use of aversive handling techniques, elevated animal stress responses, and compromised welfare outcomes (Lindahl et al., 2015; Ceballos et al., 2018). Research in abattoir settings demonstrated that handler attitudes and behaviors directly influenced physiological stress markers in cattle, with increased use of electric goads and rough handling associated with elevated cortisol concentrations (Hemsworth et al., 2011). The relationship between attitudes and safety extends to risk perception; producers who felt strong duty of care toward animals were more likely to proceed with husbandry tasks despite fatigue or environmental hazards, suggesting that positive attitudes may sometimes override safety considerations (Tone and Irwin, 2022). This complex relationship underscores the need for training that integrates welfare values with safety awareness.

Development and training of positive values and attitudes requires cognitive-behavioral interventions that address both belief systems and behavioral patterns. Cognitive-behavioral training programs have demonstrated effectiveness in improving stockperson attitudes, reducing negative handling behaviors, and enhancing animal welfare outcomes (Hemsworth et al., 2002). These interventions typically combine educational components

addressing animal behavior and welfare science with reflective exercises that encourage handlers to examine their own beliefs and practices (Ceballos et al., 2018; Vieira et al., 2023). Training effectiveness appears enhanced when programs address specific attitudinal barriers, provide opportunities for experiential learning, and include follow-up support to maintain behavioral changes over time (Panamá Arias and Špinka, 2005). Research in Brazilian dairy operations found that targeted animal welfare training significantly improved stockperson behavior, with positive interactions increasing from 43% to 61% and negative interactions decreasing from 57% to 39% (Vieira et al., 2023). The durability of these changes depends on organizational support, peer influence, and ongoing reinforcement of welfare-centered values.

Individual factors influencing values and attitudes include demographic characteristics, personality traits, cultural background, and prior experience with animals. Neuroticism has been negatively correlated with dairy herd performance and associated with higher veterinary costs, whereas conscientiousness predicted lower veterinary expenditures, suggesting that personality dimensions influence both attitudes and management outcomes (Panamá Arias and Špinka, 2005). Gender differences in attitudes have been documented, with female handlers often demonstrating stronger empathy and more positive orientations toward animal welfare (Kallioniemi et al., 2011; Cornish et al., 2018). Experience level also matters; handlers with 10 or more years of experience demonstrated better animal welfare knowledge, attitudes, and practices in Kenyan livestock markets (Wambui et al., 2018). Cultural context shapes attitudinal norms; studies comparing attitudes across different countries reveal substantial variation in beliefs about acceptable handling practices and animal welfare priorities (Boivin et al., 2007; Weyl-Feinstein et al., 2021). These individual differences highlight the importance of tailoring training approaches to address diverse starting points and cultural contexts.

Team and organizational context considerations are essential for sustaining positive values and attitudes at the farm or facility level. Organizational culture, management expectations, and peer norms create the social environment within which individual attitudes develop and persist (Napolitano et al., 2020). Research in dairy operations found that management support for welfare-centered practices and acknowledgment of stockpersons performing difficult tasks well were critical for maintaining positive attitudes toward husbandry responsibilities (Seabrook and Wilkinson, 2000). Language barriers, time constraints, and lack of training materials have been identified as organizational barriers to developing positive handler attitudes and skills (Sorge et al., 2014). The presence of welfare assessment systems and monitoring

technologies may promote adherence to best practices, though some producers express concerns about surveillance implications (Napolitano et al., 2020). Effective organizational approaches balance accountability with support, providing handlers with the resources, training, and recognition needed to maintain welfare-centered values in demanding work environments.

The importance of values and attitudes as a foundational dimension of cattle handling competence cannot be overstated. Attitudes function as the cognitive and emotional substrate from which handling behaviors emerge, influencing perception of animal cues, interpretation of cattle responses, and selection of handling techniques (Hemsworth et al., 2002; Fukasawa et al., 2017). Research consistently demonstrates that positive attitudes predict better welfare outcomes, reduced animal fear, enhanced productivity, and improved handler safety (Lensink et al., 2000a,b; Breuer, et al., 2003; Bertenshaw and Rowlinson, 2009). The malleability of attitudes through targeted training interventions offers a practical pathway for improving cattle handling systems at scale (Hemsworth, 2003; Vieira et al., 2023). However, sustaining positive attitudes requires ongoing organizational support, cultural reinforcement, and alignment between individual values and institutional practices (Seabrook and Wilkinson, 2000; Sorge et al., 2014). As the foundation of the VITAL framework, values and attitudes shape the development and expression of all other competency dimensions, making this dimension a critical target for educational interventions and policy initiatives aimed at enhancing cattle welfare and handler effectiveness.

Integration of Knowledge

Skilled cattle handling relies on the ability to integrate multiple forms of knowledge, including animal behavior, technical procedures, facility design, environmental conditions, and self-awareness, into effective handling practice. Integration of knowledge represents the cognitive dimension of cattle handling competence, encompassing understanding of cattle behavior, welfare science principles, facility design considerations, and the ability to synthesize information from multiple sources to inform handling decisions. This dimension extends beyond simple factual knowledge to include mental models of how cattle perceive and respond to their environment, comprehension of stress physiology, and application of behavioral principles to practical handling contexts (Grandin, 1987, 2003). Research demonstrates that knowledge integration enables handlers to interpret cattle behavioral signals accurately, predict animal responses to handling procedures, and adapt techniques based on contextual factors such as breed differences, prior experience, and environmental conditions (Grandin, 1987, 1999). The

cognitive integration dimension bridges theoretical understanding and practical application, providing the conceptual foundation for skilled handling performance.

Practical applications of integrated knowledge manifest in facility design decisions, handling protocol development, and real-time problem solving during cattle management procedures. Handlers with strong knowledge integration understand how cattle visual perception, flight zone dynamics, and point of balance principles can be leveraged to facilitate calm, efficient movement through handling facilities (Grandin, 1987, 1999). This understanding enables handlers to eliminate balking problems by addressing environmental factors that trigger fear responses, such as shadows, sudden movements, or visual distractions (Grandin, 1987). Research demonstrates that knowledge of cattle behavior and welfare principles translates into reduced use of aversive handling aids, decreased animal stress responses, and improved operational efficiency (Pajor et al., 2000; Progar et al., 2019). In veterinary education contexts, structured training programs that integrate behavioral science with hands-on practice have been shown to increase student competence and confidence in handling large animals (Austin et al., 2007; Hanlon et al., 2007; Stafford and Erceg, 2007). These applications illustrate how cognitive integration supports both immediate handling effectiveness and long-term skill development.

Specific components of knowledge integration include understanding of cattle sensory capabilities, comprehension of behavioral principles such as flight zones and herd dynamics, knowledge of stress physiology and welfare indicators, and familiarity with breed-specific behavioral characteristics. Knowledge of cattle sensory and behavioral characteristics, including panoramic visual fields, heightened sensitivity to high frequency sounds, and behavioral responses to facility geometry such as preference for gradual curves over sharp angles, provides a critical foundation for the design and implementation of effective handling systems (Grandin, 1987). Knowledge of flight zone principles enables handlers to position themselves strategically to encourage desired movement patterns without triggering defensive responses (Grandin, 1987; Grandin, 1999). Understanding and recognition of physiological and behavioral indicators of stress allows handlers to assess animal welfare in real time and adjust techniques accordingly (Hemsworth et al., 2011; Pajor et al., 2003). Breed differences in temperament and behavioral reactivity represent another important knowledge component; handlers who understand these variations can tailor their approach to match the behavioral characteristics of specific cattle populations (Grandin, 1987; Parham et al., 2022).

Safety and welfare implications of knowledge integration are substantial and multifaceted. Handlers with

strong behavioral knowledge are better equipped to recognize early warning signs of cattle distress or agitation, enabling proactive intervention before situations escalate to dangerous levels (Lindahl et al., 2013; Simon et al., 2016). Understanding of facility design principles based on cattle behavior reduces the likelihood of injuries caused by poor infrastructure, such as sharp corners that impede flow or inadequate lighting that creates fear-inducing shadows (Grandin, 1987, 1999). Research examining occupational health and safety in dairy operations identified inadequate training and lack of hazard awareness as primary contributors to animal-related injuries, highlighting the protective role of knowledge integration (Lindahl et al., 2012; Douphrate et al., 2013). Knowledge of appropriate handling techniques reduces reliance on aversive tools such as electric prods, which not only compromise welfare but also increase the risk of sudden, unpredictable cattle movements that endanger handlers (Waynert et al., 1999). Risk assessment frameworks that integrate knowledge of cattle behavior, facility conditions, and handler capabilities have been developed to systematically identify and mitigate safety hazards in cattle handling contexts (Geng et al., 2015).

Development and training of knowledge integration requires multimodal educational approaches that combine didactic instruction, visual demonstrations, hands-on practice, and reflective learning opportunities. Extension programs that integrate face-to-face lectures with hands-on workshops have demonstrated superior outcomes compared with lecture-only formats, with trained handlers achieving safer cattle movement patterns and reduced slipping incidents (Progar et al., 2019). Veterinary education programs that incorporate practical animal handling classes with structured assessments have successfully increased student competence and confidence in large animal handling (Austin et al., 2007; Hanlon et al., 2007; Stafford and Erceg, 2007). The effectiveness of knowledge-building interventions appears enhanced when training addresses both conceptual understanding and practical application, provides opportunities for experiential learning with immediate feedback, and includes assessment mechanisms that verify knowledge retention and transfer (Abramowicz et al., 2013; Yost et al., 2020). Working group models that facilitate peer learning through farm visits and group discussions have also shown promise for experienced producers seeking to deepen their knowledge and adopt new management practices (Vining et al., 2020).

Individual factors influencing knowledge integration include educational background, cognitive abilities, learning style preferences, and prior experience with cattle. Handlers with formal education in animal science or veterinary medicine typically demonstrate stronger foundational knowledge of behavioral and physiological

principles, though practical experience remains essential for translating theoretical knowledge into skilled performance (Austin et al., 2007; Hanlon et al., 2007). Cognitive factors such as attention to detail, pattern recognition abilities, and capacity for integrating information from multiple sources all influence how effectively handlers build and apply mental models of cattle behavior (Panamá Arias and Špinka, 2005). Experience level matters significantly; handlers with extensive cattle exposure develop richer mental models that enable rapid situation assessment and appropriate response selection (Wambui et al., 2018; Parham et al., 2022). However, experience alone does not guarantee knowledge integration; handlers may develop inaccurate mental models based on limited or biased observations, highlighting the importance of structured training that combines experiential learning with evidence-based instruction (Pajor et al., 2000).

Team and organizational context considerations shape opportunities for knowledge development and application. Organizations that prioritize training, provide access to educational resources, and create cultures of continuous learning facilitate knowledge integration among handlers at all experience levels (Abramowicz et al., 2013; Sorge et al., 2014). Peer learning opportunities, such as working groups and mentorship programs, enable knowledge transfer from experienced to novice handlers and promote collective problem-solving around handling challenges (Vining et al., 2020). Language barriers and cultural differences can impede knowledge transfer in diverse workforces, necessitating culturally appropriate training materials and multilingual educational resources (Sorge et al., 2014; Wambui et al., 2018). The availability of decision support tools, such as handling guidelines, visual aids, and assessment checklists, can scaffold knowledge application in complex or high-pressure situations (Isaacs et al., 2008). Organizational commitment to evidence-based practices, reflected in policies that align with behavioral science principles, reinforces the value of knowledge integration and supports its translation into routine handling procedures.

The significance of knowledge integration as a core dimension of cattle handling competence lies in its role as the cognitive foundation for skilled performance. Handlers cannot consistently apply appropriate techniques, recognize welfare concerns, or adapt to novel situations without robust mental models of cattle behavior and welfare (Grandin, 1987, 2003). Research demonstrates that knowledge-based interventions improve both handler behavior and animal welfare outcomes, with effects persisting over time when supported by organizational reinforcement (Hemsworth et al., 2002; Progar et al., 2019; Vieira et al., 2023). The integration of behavioral science into practical handling contexts represents a critical pathway for advancing cattle welfare at scale, as

it enables handlers to understand not only which techniques to use, but why those techniques are effective and how to adapt them to diverse situations (Grandin, 1987, 1999). As a central dimension of the VITAL framework, knowledge integration provides the cognitive foundation that enables development of technical skills, situational awareness, and adaptive learning, making it an essential target for educational interventions aimed at enhancing handler competence and animal welfare outcomes.

Technical Execution

Technical execution represents the behavioral dimension of cattle handling competence, encompassing the physical skills, movement patterns, and technique applications that handlers employ during direct interactions with cattle. This dimension includes proper positioning relative to cattle flight zones and points of balance, appropriate use of handling aids, timing and coordination of movements, and physical techniques for guiding cattle through facilities or performing husbandry procedures (Grandin, 1987). Research demonstrates that proficient technical execution minimizes stress, improves cattle flow, and reduces risk to both animals and handlers (Petherick et al., 2009a,b; Progar et al., 2019). Technical skills develop through deliberate practice with feedback, progressing from conscious, effortful performance to fluid, automatic execution as expertise develops (Ericsson et al., 1993). The technical execution dimension represents the observable manifestation of handler competence, translating knowledge and attitudes into effective action.

Practical applications of technical execution span routine handling procedures, facility management, emergency responses, and specialized husbandry tasks. Skilled handlers demonstrate precise positioning that leverages cattle flight zones to encourage the desired movement without triggering fear responses, maintaining appropriate pressure and release patterns that facilitate calm, efficient flow (Grandin, 1987, 1999; Rennie et al., 2003; Destrez et al., 2018). Research examining natural stockmanship training found that handlers trained in obtaining submissive responses through proper positioning and timing achieved reduced avoidance reactions and improved cattle calmness compared with conventional handling approaches (Abramowicz et al., 2013). Technical proficiency in halter training, restraint procedures, and movement through handling facilities directly influences animal stress levels, with gentle, consistent techniques being associated with reduced physiological stress markers (Breuer, et al., 2000; Lensink et al., 2000a). In veterinary contexts, technical competence in large animal handling is essential for safe examination and treatment procedures; structured training programs have successfully improved veterinary students' handling skills and

confidence through practical classes and performance assessments (Austin et al., 2007; Hanlon et al., 2007; Stafford and Erceg, 2007).

Specific components of technical execution include flight zone management, point of balance utilization, appropriate handling aid use, body positioning and movement coordination, and procedure-specific techniques. Flight zone management requires handlers to understand the invisible boundary around each animal that triggers movement when crossed, adjusting their position and approach speed to maintain optimal pressure (Grandin, 1987, 1999). Point of balance principles, such as recognizing that positioning forward of the shoulder can slow or stop cattle movement and positioning behind the shoulder encourages forward movement, enable handlers to direct cattle movement efficiently (Grandin, 1987). Handling aid use encompasses both appropriate tools (such as flags or paddles for visual guidance) and avoidance of aversive implements; research demonstrates that electric prod use increases stress responses and should be minimized or eliminated (Waynert et al., 1999; Hemsworth et al., 2003, 2011; Tone and Irwin, 2022). Body positioning and movement coordination involve smooth, deliberate actions that avoid sudden movements or loud vocalizations that trigger fear responses (Waynert et al., 1999). Procedure-specific techniques, such as proper halter fitting, safe restraint methods, and gentle touching during examinations, require targeted skill development through supervised practice (Lensink et al., 2000a,b; Abramowicz et al., 2013).

Safety and welfare implications of technical execution are immediate and consequential. Proficient technical skills reduce the likelihood of cattle injuries caused by rough handling, inappropriate restraint, or facility-related accidents (Petherick et al., 2009a,b; Douphrate et al., 2013). Handler safety is directly influenced by technical competence; skilled positioning and movement patterns minimize exposure to kicks, charges, or crushing injuries that may occur when cattle feel threatened or cornered (Lindahl et al., 2012, 2013). Research examining dairy cattle handling found that handlers trained through combined lecture and hands-on workshop formats achieved significantly reduced cattle slipping incidents and improved walking behavior, demonstrating the safety benefits of technical skill development (Progar et al., 2019). Conversely, poor technical execution, including excessive prod use, shouting, and rough physical contact, elevates animal stress, increases fear responses, and compromises both welfare and productivity outcomes (Hemsworth et al., 2011; Lindahl et al., 2015; Ceballos et al., 2018). The quality of technical execution during routine procedures has lasting effects on cattle temperament; research demonstrates that consistent gentle handling reduces fear of humans over time, whereas poor handling

quality sustains or exacerbates fearfulness (Petherick et al., 2009a,b). These findings underscore the critical importance of technical skill development for achieving both safety and welfare objectives.

Development and training of technical execution requires hands-on practice with immediate feedback, progressive skill building, and opportunities for deliberate refinement. Effective training programs combine demonstration of proper techniques, supervised practice with live animals, and structured feedback on performance (Abramowicz et al., 2013; Progar et al., 2019; Vieira et al., 2023). Extension programs that include hands-on workshop components demonstrate superior outcomes compared with lecture-only formats, with participants showing improved technical skills and sustained behavior change (Progar et al., 2019; Vining et al., 2020). Natural stockmanship training approaches that emphasize reading cattle body language and adjusting handler positioning accordingly have shown effectiveness in reducing avoidance reactions and improving cattle calmness (Abramowicz et al., 2013). Veterinary education programs that incorporate practical handling classes with performance assessments successfully develop technical competence and confidence in students (Austin et al., 2007; Hanlon et al., 2007; Progar et al., 2019). The progression from novice to expert technical performance requires extensive practice; research suggests that handlers benefit from ongoing skill refinement opportunities and periodic retraining to maintain proficiency and prevent skill degradation (Vieira et al., 2023).

Individual factors influencing technical execution include physical capabilities, motor coordination, prior experience with cattle, and capacity for integrating feedback. Physical attributes such as strength, agility, and coordination influence handlers' ability to perform certain techniques safely and effectively, though proper technique can often compensate for physical limitations (Kallioniemi et al., 2011). Prior experience with cattle provides familiarity with animal responses and develops the timing and positioning skills essential for proficient handling (Wambui et al., 2018; Parham et al., 2022). Individual differences in motor learning abilities affect the rate at which handlers acquire technical skills; some individuals demonstrate rapid skill acquisition, whereas others require more extensive practice to achieve proficiency (Ericsson et al., 1993). The capacity to integrate feedback from instructors and from cattle behavioral responses is critical for skill refinement; handlers who attend to animal reactions and adjust their techniques accordingly develop proficiency more rapidly than those who persist with ineffective approaches (Abramowicz et al., 2013; Vieira et al., 2023). Confidence also matters; handlers who feel uncertain about their technical abilities may hesitate or use excessive force, both of which

compromise handling effectiveness (Austin et al., 2007; Hanlon et al., 2007).

Team and organizational context considerations shape opportunities for technical skill development and maintenance. Organizations that provide structured training programs, allocate time for skill practice, and create cultures that value technical proficiency facilitate skill development among handlers (Sorge et al., 2014; Progar et al., 2019). Peer modeling and mentorship enable novice handlers to observe and learn from experienced colleagues, accelerating skill acquisition through social learning mechanisms (Vining et al., 2020). Facility design and equipment quality influence technical execution; well-designed facilities that incorporate behavioral principles reduce the technical demands on handlers and enable more effective performance (Grandin, 1987, 1999). Organizational policies regarding handling aid use, such as restrictions on electric prod use, shape the technical repertoire that handlers develop and reinforce welfare-centered skill sets (Waynert et al., 1999; Hemsworth et al., 2011). Performance monitoring and feedback systems, including tools such as the Stockman's Scorecard, provide objective assessment of technical execution quality and identify areas for improvement (Yost et al., 2020). Organizational commitment to ongoing training and skill maintenance ensures that technical proficiency is sustained over time rather than degrading due to lack of practice or reinforcement.

The importance of technical execution as a core dimension of cattle handling competence lies in its direct impact on animal welfare, handler safety, and operational efficiency. Technical skills represent the behavioral interface between handler and animal, determining the immediate quality of each interaction and shaping long-term cattle temperament and human-animal relationship quality (Petherick et al., 2009a,b). Several studies have documented relationships between handler technical proficiency and reduced stress responses in cattle, with skilled execution of handling procedures associated with lower cortisol concentrations, decreased fear behaviors, and improved measures of production parameters (Petherick et al., 2009b; Abramowicz et al., 2013; Progar et al., 2019). The ability to teach technical skills through structured, hands-on programs offers a practical pathway for improving cattle handling systems, with benefits evident in both immediate handling outcomes and sustained behavioral changes (Progar et al., 2019; Vieira et al., 2023). However, technical proficiency alone is insufficient without the attitudinal foundation and cognitive understanding provided by other VITAL dimensions; skilled technique application requires knowing when and why to use specific approaches, not just how to perform them (Grandin, 1987, 1999). As a central dimension of the VITAL framework, technical execution translates

knowledge and values into observable action, making it an essential target for training interventions and a critical indicator of overall handler competence.

Maintaining high technical standards requires ongoing training, feedback, and organizational support. Overwork, time pressure, or limited supervision can erode execution quality, whereas operations that invest in regular assessment and refresher training demonstrate greater consistency and safer outcomes (Grandin, 2003; Isaacs et al., 2008). Programs such as the Work Crew Performance Model identify high-impact tasks and target them for focused improvement (Isaacs et al., 2008). Technical execution is therefore not a static achievement but an ongoing process of refinement. As the physical and procedural foundation of cattle handling skill, this dimension enables precise, humane, and efficient interaction with cattle and serves as a measurable contributor to animal welfare, handler safety, and operational efficiency.

Situational Awareness

Situational awareness (SA) is a core perceptual and cognitive skill in cattle handling, referring to a handler's ability to monitor animals, facilities, environmental conditions, and team dynamics in real time. Situational awareness has been defined as the perception of entities in the environment, comprehension of their meaning, and projection of their status in the near future (Endsley, 2021; Munir, et al., 2022). In expertise research, SA is considered a critical cornerstone of performance across domains, from aviation to medicine, developing alongside and supporting more observable features such as physical skill and decision making (Endsley, 2021). Successful performance in complex, dynamic environments depends on such domain-dependent capacities, which are supported by domain-independent cognitive abilities, including perception, memory, attention, and executive control (O'Brien and O'Hare, 2007). In the context of cattle handling, SA therefore requires continuous scanning of the environment, interpretation of both subtle and overt animal signals, and timely adjustment of behavior to maintain safety, efficiency, and animal welfare.

Handlers with strong SA consistently monitor their position relative to cattle, recognize early indicators of fear, agitation, or aggression, and identify environmental risks such as slippery surfaces, visual distractions, or malfunctioning equipment (Grandin, 1987; Breuer et al., 2000). Importantly, this dimension also includes self-awareness: the ability to recognize how one's own posture, movement, emotional state, and pace influence animal responses (Ellingsen et al., 2014). Skilled handlers regulate their body position, movement speed, and voice use deliberately, adjusting their approach in real time to prevent escalation. Calm posture, controlled

motion, and a steady voice are outward expressions of this internal awareness. Emerging work from high-risk operational domains suggests that the capacity to develop SA may be moderated by individual differences in mindfulness and emotional self-regulation (Büssing et al., 2013). The concept of "mind fitness," for example, proposes that the ability to remain present and regulate emotional responses enhances performance in critical, high stress environments by improving adaptive reaction, resilience, and recovery (Stanley et al., 2011). Similarly, awareness training in military personnel has been shown to strengthen the ability to perceive oneself in relation to the surrounding environment, a foundational component of SA (Strozzi-Heckler, 2011). Although developed outside of livestock systems, these findings suggest that internal regulatory capacities may represent an underexplored factor in cattle handling skill, warranting further study to determine how SA can be measured, supported, and intentionally trained.

Spatial awareness is a fundamental component of SA. Proficient handlers understand animal blind spots, flight zones, and points of balance, and they use this knowledge to guide cattle smoothly and predictably. They approach animals at strategic angles and adapt techniques based on production context, recognizing, for example, that dairy cattle with frequent human contact generally exhibit smaller flight zones than extensively managed beef cattle. This awareness extends to facilities and equipment, with experienced handlers evaluating handling areas from the animal's perspective, or at their eye level, to identify potential distractions such as shadows, dangling chains, or objects left in walkways that may cause balking or hesitation (Grandin, 2009).

Situational awareness also underpins risk prevention and occupational safety. Many cattle handling injuries arise from lapses in awareness, including improper positioning, misjudgment of animal intent, or failure to maintain escape routes (Doughrte et al., 2013). Handlers with high SA anticipate sudden movements, recognize hazardous zones, and maintain spatial buffers that reduce the likelihood of kicks, crushing injuries, or trampling. Assessment tools such as the Stockman's Scorecard incorporate SA through criteria related to pen fill rates, avoidance of blind spots, and understanding of facility flow and animal movement patterns (Yost et al., 2020).

At an advanced level, SA supports behavioral prediction. Experienced handlers detect early warning signs of stress or aggression such as tail elevation, pawing, pinned ears, or changes in movement speed, and adjust pressure, positioning, or pacing before problems occur. This anticipatory capacity is particularly important in high-risk or unfamiliar contexts, including working with bulls, separating cows and calves, or introducing new animals to a group. The ability to interpret intent

and act proactively, rather than reactively, is central to safe and humane handling.

The development of SA appears to be both experiential and trainable. Although individual differences in attention, perception, and emotional regulation may influence baseline capacity (O'Brien and O'Hare, 2007), research in stockperson training demonstrates that structured, hands-on instruction improves handlers' behavioral consistency, animal responses, and safety outcomes (Hemsworth et al., 1996, 2002; Hemsworth, 2003). Training approaches that incorporate guided observation, facility walkthroughs, and real-time feedback are conceptually aligned with SA theory, which holds that perception and projection skills must be practiced in dynamic contexts rather than acquired through passive instruction alone (Endsley, 2021). Accordingly, approaches that embed learners in live environments may better support the development of SA and decision making in dynamic environments. Endsley and Robertson (2000) suggest that training focused on improving the cognitive skills thought to underpin SA (e.g., attention sharing, contingency planning) should result in an increased ability to maintain SA.

Individual factors such as experience, maintaining attention and focus, and fatigue are likely to influence SA. Overconfidence or complacency sometimes observed among experienced handlers may contribute to reduced vigilance, particularly during routine tasks where risk may be underestimated. Conversely, positive engagement with cattle and reflective practice are associated with greater attentiveness to animal signals. Situational awareness must therefore be actively maintained, especially under time pressure or changing conditions.

Situational awareness also influences teamwork and communication. In multiperson handling scenarios, shared awareness of animal location, movement, and human positioning is essential. Effective handlers monitor not only cattle behavior but also the actions and placement of team members, communicating clearly and adjusting roles as needed to maintain safety and consistency.

The welfare implications of SA are well documented. Early detection of distress allows handlers to modify their approach before fear or agitation escalates, reducing the likelihood of chronic stress responses and injury. Animals that experience calm, predictable handling are easier to manage and less likely to develop avoidance behaviors, reinforcing positive human-animal interactions. Similarly, many occupational injuries follow predictable patterns: getting caught between animals and barriers, misjudging movement trajectories, or failing to recognize flight or defensive behavior. Keen awareness and the ability to read and interpret behavioral cues requires attentional discipline, which is necessary for hazard recognition.

Although facility design can support SA, infrastructure alone cannot replace attentive handling. Even well-designed systems depend on alert, responsive handlers to prevent incidents. Conversely, strong SA can partially compensate for suboptimal facilities by enabling handlers to adapt positioning and pacing to reduce risk.

Situational awareness in cattle handling is not a fixed trait, but a learned and practiced skill. It enables handlers to respond effectively in complex, changing environments, protect themselves and others, and support better welfare and productivity outcomes. As livestock systems increasingly emphasize safety, welfare, and efficiency, SA warrants formal recognition, training investment, and assessment alongside other core competencies of stockmanship.

Commitment to Continuous Learning

Cattle handling proficiency is not static, but develops over time through intentional learning, reflection, and adaptation. This dimension captures a handler's sustained engagement in refining practice in response to new information, feedback, and experience. Rather than a one-time acquisition of skill, continuous learning reflects an ongoing orientation toward improvement that supports long-term consistency, safety, and animal welfare.

Formal training programs consistently demonstrate positive impacts on stockperson behavior and cattle outcomes. Handlers who participate in targeted cognitive-behavioral interventions report greater confidence, calmer demeanor, and reduced reliance on aversive techniques (Arney, 2009; Heinsius et al., 2024; Noffsinger and Morrissey, 2024; Vieira et al., 2023). These changes are observable in practice: trained stockpeople engage more frequently in positive tactile interactions (Schmitz et al., 2020; Lomb et al., 2021) and rely less on forceful handling methods (Leon et al., 2020), with corresponding reductions in cattle fear responses such as increased flight distance or agitation during milking (des Roches et al., 2016; Rault et al., 2020). Evidence further suggests that training effects are durable, with improvements in handler behavior and animal responses persisting for months following intervention (Hemsworth et al., 2002; Napolitano et al., 2020; Fernandes et al., 2021).

Beyond formal instruction, experiential learning plays a critical role in skill development. Often described as "learning by doing," this process emerges through repeated observation, trial and error, and reflection on animal responses across handling contexts. Skilled stockpeople emphasize the importance of developing the ability to read animals, anticipate movement, and adjust their own behavior accordingly (Hemsworth et al., 1989). Swedish dairy farmers, for example, described proficiency in cattle handling as accumulating experience over time:

recognizing patterns, developing intuition, and refining technique (Lindahl et al., 2012). This process involves both physical skill and emotional regulation, as animals often mirror handler calmness or stress (Kallioniemi et al., 2011). In this way, experience reinforces technical execution while strengthening the self-awareness needed to sustain safe, low-stress interactions.

Social learning and team culture further shape continuous improvement. Handlers frequently report adopting better techniques through observing others, discussing animal behavior with peers, and reflecting on shared experiences (Yost et al., 2020). Farms with close contact to trained individuals often demonstrate intermediate improvements in handling quality, even among workers without direct training exposure, illustrating how knowledge diffuses through informal networks and shared standards (Ceballos et al., 2018). When management fosters open communication and encourages discussion of cattle behavior, a culture of curiosity and accountability supports sustained improvement (Bassi et al., 2019).

Assessment frameworks such as the Stockman's Scorecard (Yost et al., 2020) and the Working Environment Screening Tool in Agriculture risk assessment (Geng et al., 2015) and other tools for assessing stockperson behavior provide structured opportunities for self-evaluation and feedback (Destrez et al., 2018). These tools identify common handling deficits (Rennie et al., 2003) such as overfilling crowd pens, working in blind spots, or failing to adjust pressure, thereby allowing handlers to make targeted changes. However, improvement depends not only on audit results, but also on a willingness to reflect, adapt, and engage in ongoing goal setting (Austin et al., 2007; Rault et al., 2020).

Despite the recognized benefits of training and assessment, barriers to continuous learning persist. Economic stress, staffing shortages, time pressure, and fatigue can reduce capacity for skill refinement (Isaacs et al., 2008). Some producers remain more inclined to invest in physical infrastructure than in behavioral training, even when the latter offers clearer welfare and productivity returns (Grandin, 2003). Even highly motivated handlers may lack awareness of key concepts such as pain masking, habituation, or animal learning if educational materials are not routinely updated or accessible (Weyl-Feinstein et al., 2021).

Individual and psychosocial characteristics also influence engagement with learning. Female handlers often report stronger emotional connection with animals and greater concern for animal distress, which may enhance attentiveness to feedback and openness to skill development (Cornish et al., 2018; Weyl-Feinstein et al., 2021). Experience level further shapes learning dynamics: although new handlers may approach training with enthusiasm, some experienced workers resist change, par-

ticularly when training is perceived as remedial rather than developmental (Vieira et al., 2023). Conversely, handlers who report satisfaction in their work and interest in skill building tend to demonstrate safer handling practices and more consistent animal outcomes (Kallioniemi et al., 2011).

Ultimately, commitment to continuous learning reflects a sustained capacity to refine practices over time. Skilled cattle handlers are not simply trained; they remain engaged in learning through observation, feedback, and reflection, adjusting their behavior in response to animal cues and changing conditions. Within the VITAL model, this dimension supports and integrates all others by enabling adaptation, consistency, and improvement. As such, lifelong learning supports continuous improvements in animal welfare, worker safety, and workforce retention.

Taxonomy of Handler Skill

Building from these dimensions, we developed a handler-centered taxonomy of cattle handling skill, termed the VITAL model, which integrates 5 related dimensions: Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Continuous Learning (Table 1; Figure 2). This model is intended to capture both observable handling actions and underlying perceptual and cognitive components.

DISCUSSION

Context and Prior Work

A substantial body of research now demonstrates that individuals working directly with cattle can profoundly influence animal welfare outcomes. Stockperson qualities, including attitudes, empathy, and confidence, have measurable effects on cattle stress, fear, and productivity, underlining the pivotal role of human-animal interactions in agricultural systems (Moggy et al., 2017; Ebingerhaus et al., 2020). The impact of these interactions is supported by evidence across multiple species, revealing that handler attitudes and behaviors correlate strongly with animal welfare, health, and even the efficiency of production systems (Leon et al., 2020; Pol et al., 2021).

Despite this, much of the scientific literature has examined human factors in isolation rather than recognizing stockmanship as a suite of interrelated skills within broader farming practices and environments (Colditz et al., 2014). More recent approaches advocate for integrated management, accounting for how training, housing, animal handling protocols, and social and work environments interact to shape welfare outcomes (Simova et al., 2016). Objective assessment

Table 1. The VITAL model: A taxonomy of cattle handling skill¹

Dimension	Action	Characteristic
Values and Attitudes (cognitive/affective)	Demonstrates empathy, patience, and a sense of responsibility for cattle; recognizes animals as sentient individuals	Reflects a welfare-oriented approach to cattle care, expressed through calm, patient, and humane handling practices
Integration of Knowledge (cognitive)	Integrates knowledge of cattle behavior, physiology, and environment to guide handling decisions	Reflects an ability to synthesize multiple knowledge sources and apply judgment when adapting handling practices across changing conditions
Technical Execution ² (motor/psychomotor)	Executes handling techniques with consistency, precision, and appropriate timing	Reflects a high level of technical proficiency, characterized by controlled, consistent execution of handling techniques
Situational Awareness (perceptual/cognitive)	Maintains awareness of cattle behavior, body position, and environmental conditions during handling	Reflects a habit of close observation and attentiveness to cattle and surrounding conditions during handling
Commitment to Continuous Learning ³ (reflective/affective)	Seeks feedback, learns through experience, and engages in continuing education to refine handling practices over time	Reflects a willingness to learn from experience and others, incorporating feedback and continuing education into ongoing practice refinement

¹The model describes 5 interrelated dimensions of cattle handling skill. Actions represent observable expressions of skilled handling in specific situations, whereas characteristics reflect stable capacities or tendencies evident through consistent handling across situations and time. Dimensions are conceptually related but nonhierarchical. Footnotes clarify construct boundaries where dimensions may otherwise appear overlapping.

²Technical execution refers to the physical performance of handling techniques, independent of the cognitive processes used to select or adapt those techniques.

³Continuous learning refers to observable engagement with experiential learning, feedback, and continuing education, rather than inherent motivation or personality traits.

tools and evidence-based training programs have been recommended to facilitate these advances (Hemsworth, 2007; Grandin, 2018), and studies show that effective interventions improve not only animal welfare, but also stockperson satisfaction and job performance (Waiblinger et al., 2002; Pol et al., 2021).

At the conceptual level, established animal welfare frameworks such as the Five Freedoms, Five Domains, and Three Circles provide structured approaches for describing what animals experience, express, and require to experience good welfare. These frameworks appropriately center on animal-based outcomes, particularly affective states and their biological and environmental determinants. Notably, the 2020 update of the Five Domains Model explicitly incorporates human–animal interactions within domain 4, detailing how human attributes and behaviors may generate negative or positive affective impacts in animals (Mellor et al., 2020). Mounting evidence demonstrates that human actions and interactions particularly during active handling, are key inputs shaping those animal experiences (Fleming et al., 2016; Cole and Fraser, 2018; Nawroth et al., 2019). Although contemporary welfare frameworks now recognize and structure the welfare impacts of human–animal interactions, the competencies, perceptual skills, and cognitive attributes of handlers that give rise to those interactions remain less systematically conceptualized. As a result, although welfare models effectively define animal outcomes, the multidimensional construct of handler skill that influences those outcomes remains comparatively underdeveloped in the literature (Hemsworth, 2007; Balzani and Hanlon, 2020; Temple and Manteca, 2020).

The current understanding thus supports moving toward more comprehensive animal welfare strategies that account for stockperson competencies and management culture alongside traditional animal-centered metrics. This broader perspective is essential for achieving sustainable improvements in both animal welfare and enterprise outcomes (Grandin, 2009, 2018; Phillips et al., 2009; IFO, 2014).

Taxonomy Overview and Novelty

The VITAL model organizes the foundational components of cattle handler skill into 5 dimensions. Together, they consolidate diverse empirical findings into a unified structure that illustrates how human attributes directly shape animal welfare.

Each dimension captures critical components of handler skill that contribute to stockmanship: the values and attitudes that guide behavior toward animals; the integration of knowledge that connects understanding of cattle behavior, physiology, and environment; the technical execution of active handling practices; the SA that allows handlers to perceive and adapt in real time; and the commitment to continuous learning that supports continuous improvement over time. Unlike work focused on single variables, the VITAL model integrates cognitive, behavioral, emotional, and adaptive dimensions into a cohesive structure.

When considered alongside established animal welfare frameworks such as the Three Circles (Fraser et al., 1997), cattle handling can be understood as a form of human–animal interaction with direct implications for

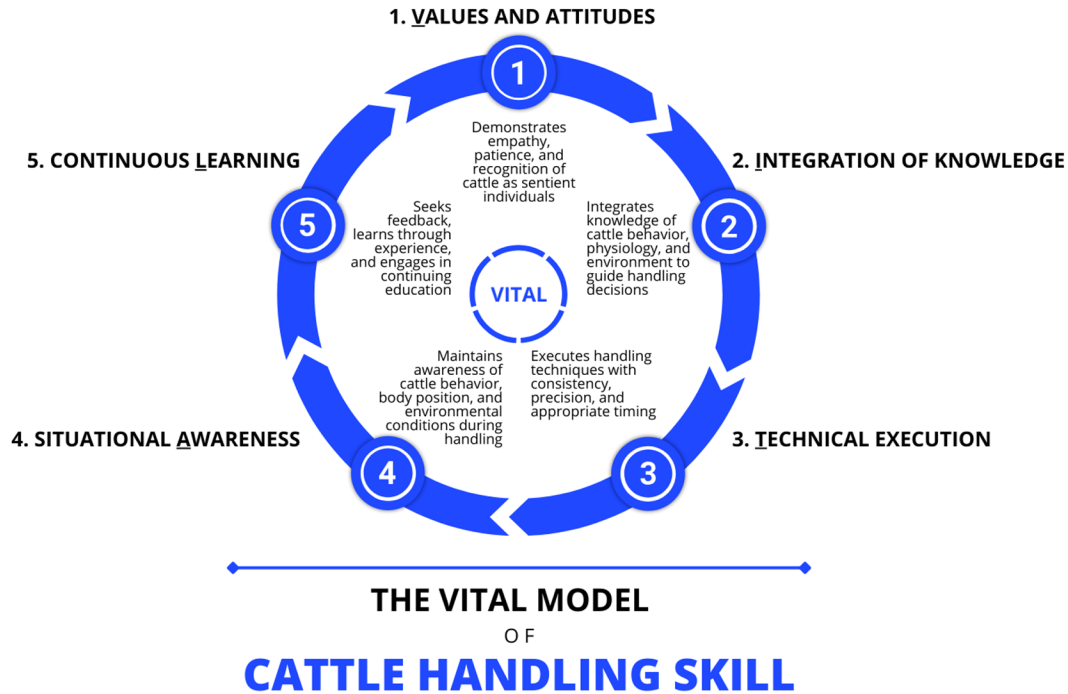


Figure 2. The VITAL model of cattle handling skill. The VITAL model depicts 5 interrelated dimensions of handler skill: Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Continuous Learning. The figure highlights observable handling actions associated with each dimension. Together, these dimensions describe skilled cattle handling as an integrated practice grounded in observation, experience, and evidence-based action.

animal welfare outcomes. Handling practices influence cattle affective states through learned fear or calm responses to human interaction, restraint, and movement, and can affect biological functioning through stress-related physiological responses and downstream impacts on productivity and health. In this sense, handler skill and handling practices are proximal determinants of welfare within livestock systems, particularly under conditions of domestication where animals' experiences are shaped by routine human interaction.

Implications for Training, Practice, and Policy

The VITAL model provides a foundation for improving training, assessment, and workplace culture related to cattle handling. By framing handler skill as a set of interrelated competencies, it highlights where education, evaluation, and organizational support can have the greatest impact.

- **Training and education:** Instructional programs can be structured around the VITAL model to balance technical proficiency with empathy, decision making, and reflection. Addressing both the physical and cognitive aspects of handling may better prepare handlers to work with confidence and competence across a range of real-world conditions.
- **Assessment and feedback:** The VITAL model offers a conceptual basis for developing practical, evidence-based tools to evaluate handler skill. Future assessment systems should capture growth over time, integrating observable behavior with self-reflection and situational judgment.
- **Workplace culture:** Recognizing values and attitudes as part of professional practice for handlers underscores the need for organizational environments that reinforce empathy, patience, respect toward livestock, and openness to feedback. Although “time is money” in production settings, effective cattle handling often requires slowing down to prevent stress, slips, or injuries that ultimately cost more in time and resources. When supervisors communicate their support of this principle (practices such as pausing to let cattle settle or using quiet cues instead of shouting or hitting) it reinforces that compassion, patience, awareness, and willingness to learn are valued hallmarks of professionalism.
- **Safety and risk management:** Building SA through experience, guided observation, and feedback helps handlers anticipate animal movement, recognize fatigue, and prevent injury. Emphasizing these perceptual and decision-making skills supports safer,

low-stress cattle handling that may reduce risk of physical injury for workers.

- **Policy and industry credibility:** As public scrutiny grows and animal welfare is increasingly viewed as a reflection of professional accountability, integrating the VITAL model into training standards, certification programs, and audit systems can help sustain social license and demonstrate continuous improvement in the cattle sector. National programs such as the Farmers Assuring Responsible Management (FARM) Animal Care Program (NMPF, 2024), which include expectations for annual continuing education in stockmanship and animal care, provide a relevant example of how handler development can be systematically documented and supported. By explicitly connecting handler development to animal outcomes, the VITAL model positions human capability as a central driver of welfare progress, workforce sustainability, and the preservation of social license in modern livestock systems.

Strengths and Limitations

This review integrates peer-reviewed literature to develop a cohesive, evidence-based taxonomy of cattle handler skill. A key strength lies in its grounded approach: the dimensions emerged inductively from the data rather than being imposed a priori, ensuring the framework reflects patterns documented across diverse production systems and methodologies. The inclusion of studies spanning multiple countries, disciplines, and handling contexts enhances the framework's relevance and transferability.

However, the available literature presents several limitations. Study quality and reporting standards varied considerably, particularly in older publications. Few studies explicitly aimed to define or objectively measure handler skill, so synthesis required interpretive analysis. Restricting the search to English-language publications within WOS may have excluded relevant work in other languages or databases.

Future Directions

Future research should translate the VITAL model into practical assessment and training tools that can be applied across livestock sectors. Longitudinal studies could examine how handler competencies develop and interact over time, from beginner through expert, whereas experimental interventions could test whether targeted training in specific dimensions improves both handler performance and animal outcomes.

Cross-cultural and system-level comparisons would further clarify how values, attitudes, and learning prac-

tices vary among production contexts, regions, and organizational structures. Incorporating physiological and behavioral welfare indicators alongside human performance metrics could strengthen causal understanding of how handler skill affects cattle welfare. Finally, integrating these measures into audit systems, credentialing programs, and workforce training pipelines would enable the VITAL model to guide not only scientific inquiry, but also the professionalization of livestock care.

CONCLUSIONS

This systematic review identified and synthesized peer-reviewed studies to develop the VITAL model, a 5-theme taxonomy describing the core competencies of cattle handler skill: Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Continuous Learning. The framework positions handler competence as a proximal determinant of animal welfare; one that directly influences cattle behavior, stress, and safety outcomes for both humans and animals. By aligning human attributes with recognized welfare frameworks such as the Three Circles model, the VITAL model bridges the human and animal dimensions of livestock care. It provides both a conceptual and practical foundation for assessment, training, and policy, recognizing animal welfare depends not only on system design, environment, and resources, but also on the skill and judgment of the people who work with animals every day. Ultimately, recognizing and developing these competencies is essential to improving welfare outcomes, supporting workforce sustainability, and maintaining social license in modern cattle production.

NOTES

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nary screening of record titles prior to data extraction. The tool was used solely to support identification of potentially relevant records; all inclusion and exclusion decisions were independently reviewed and finalized by the authors. The authors take full responsibility for the content of the publication. A PRISMA protocol was not prepared, and this systematic review was not prospectively registered. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AI = artificial intelligence; Misc = miscellaneous; PRISMA = Preferred Reporting Items for Systematic reviews and Meta-Analyses; SA = situational awareness; VITAL = Values and Attitudes, Integration of Knowledge, Technical Execution, Situational Awareness, and Commitment to Continuous Learning; WOS = Web of Science.

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